



TRANSORG AGENTIQ DATATHON • TRACK 3

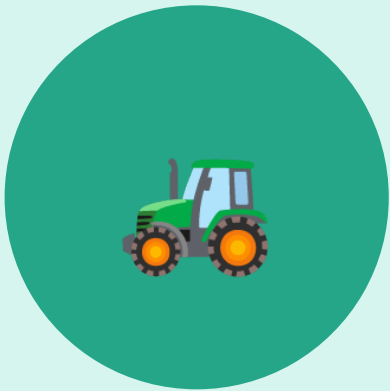
AgriTech: Mandi-to-Market Supply Chain Optimizer

A reproducible data-rescue pipeline, an executive Power BI dashboard, and a text-to-chart AI agent — built on real mandi, price, transport and weather data.

Presented for: TransOrg Analytics — AgentIQ Datathon

Section: Track 3 Submission

github.com/rohansaii/AgriTech-Mandi-to-Market-Supply-Chain-Optimizer



Agriculture's data problem is a decision problem.

Introduction

Agricultural supply chains generate large volumes of data across mandis, prices, transport and weather — but it's rarely clean enough to act on.

The Challenge

Mandi IDs, crop names, units and prices arrive in inconsistent, multilingual formats across 5 separate source files — making even basic aggregation unreliable.

Our Solution

A documented cleaning pipeline turns 5 messy raw files into 5 query-ready, star-schema CSVs — with a Power BI dashboard and AI agent on top.

Problem Statement

Farmers, traders and mandi boards face a common dilemma: the raw data that could guide decisions is too fragmented to trust.



Existing raw exports mix ID formats, languages, units and currency notations — making decision-making incomplete and error-prone.

Raw & Fragmented

-  **5 mandi ID formats**
MANDI001, M017, 034...
-  **3 languages for crops**
English, Hindi, Punjabi
-  **4 currency notations**
₹1,234 / Rs.1,234 / 1234/-

Clean & Analytics-Ready

-  **1 canonical ID format**
MANDI0## everywhere
-  **1 mapped crop taxonomy**
raw value preserved for audit
-  **1 numeric price field**
symbols & commas stripped



Motivation

The motivation behind this project is to close the gap between raw agricultural records and decisions people can actually trust.



Data-Driven Decisions

Give mandi boards and traders a single, trustworthy view of arrivals, prices and logistics instead of scattered exports.



Reduce Guesswork

Replace manual spreadsheet reconciliation with a pipeline that flags every assumption it makes.



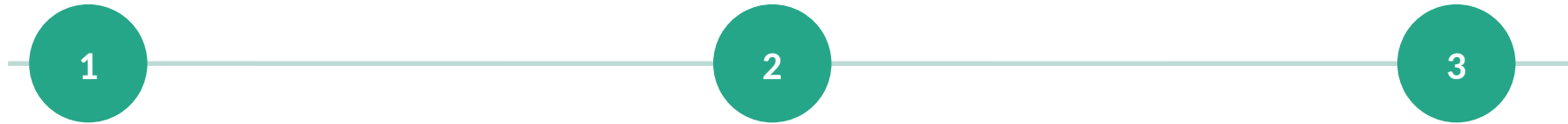
Smarter Agri-Commerce

Make it easier to catch price crashes, transport delays and weather risk before they compound.



Objectives

The main objectives of this project:



Rescue & Standardize

Clean 5 raw source files into consistent, documented, analytics-ready data.

Build the Analytics Layer

Model a star schema with clear metrics: arrivals, price vs. MSP, transit, weather.

Enable Self-Serve Insight

Ship an executive dashboard and a natural-language agent for instant answers.



Key Features

This project includes several features built for auditability and reuse:



Reproducible Pipeline

Idempotent Python cleaning script — rerun anytime with the same result.



Auditable Cleaning Rules

Every transformation is documented with its reasoning in the data dictionary.



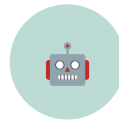
Star-Schema Analytics

1 dimension + 4 fact tables, ready for BI tools or SQL.



Executive Dashboard

5-page Power BI report covering overview, price, logistics and weather.



Text-to-Chart Agent

Ask questions in plain English, get the right chart instantly.



Flagged Assumptions

Every imputed or corrected value is flagged, never silently changed.



System Overview

The system takes data from multiple sources, processes it, and uses a documented pipeline to produce a trustworthy analytics layer and dashboard.



Architecture Diagram

The system architecture is layered: each layer reads the output of the one before it, so failures stay isolated and traceable.



Reads the raw files, regenerates the five clean CSVs, and every run produces the same output — the pipeline is idempotent by design.

`docs/star_schema_and_powerbi_guide.md`

Documents table relationships, DAX measures and dashboard design decisions for full reproducibility.

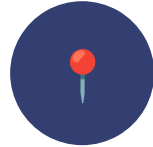
Data Sources

The system is built from five raw Track 3 datathon files, each covering a different domain:



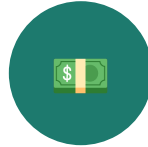
Mandi Arrivals

Daily crop arrival quantities per
mandi



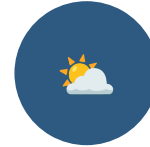
Mandi Master

Canonical mandi identifiers,
names & states



Price & MSP

Modal prices vs. minimum
support price, JSON



Weather Sensors

Temperature & rainfall telemetry,
Excel



Transport Logistics

Vehicle trips, distance & delay
records



Data Preprocessing

Raw data collected from many sources is often noisy and incomplete. Therefore, preprocessing is required to clean, standardize and validate every field.

This ensures the analytics layer and dashboard are built on reliable, trustworthy inputs.

→ Clean the Data

Remove noise and inconsistencies from raw sensor and API feeds.

→ Handle Missing Values

Impute or remove incomplete records to ensure data integrity.

→ Normalize Units

Scale variables so every unit is compatible for modeling.

→ Extract Flags

Surface flagged, corrected or imputed fields for auditability.



Trade-Off: Speed vs. Rigor in Cleaning

There's always a trade-off between how fast a cleaning rule ships and how much judgment it encodes. This pipeline documents that trade-off instead of hiding it.

AUTOMATIC RULES

Fast, deterministic, zero ambiguity

- Currency symbol & comma stripping
- Unit conversion (Tonnes/KG → Quintals)
- Timestamp standardization to IST
- Duplicate record removal

JUDGMENT CALLS

Slower, documented, reviewable

- Rice vs. Paddy crop-name mapping
- Assuming missing distance units = km
- Median imputation for farmer counts
- No sensor-to-mandi weather mapping

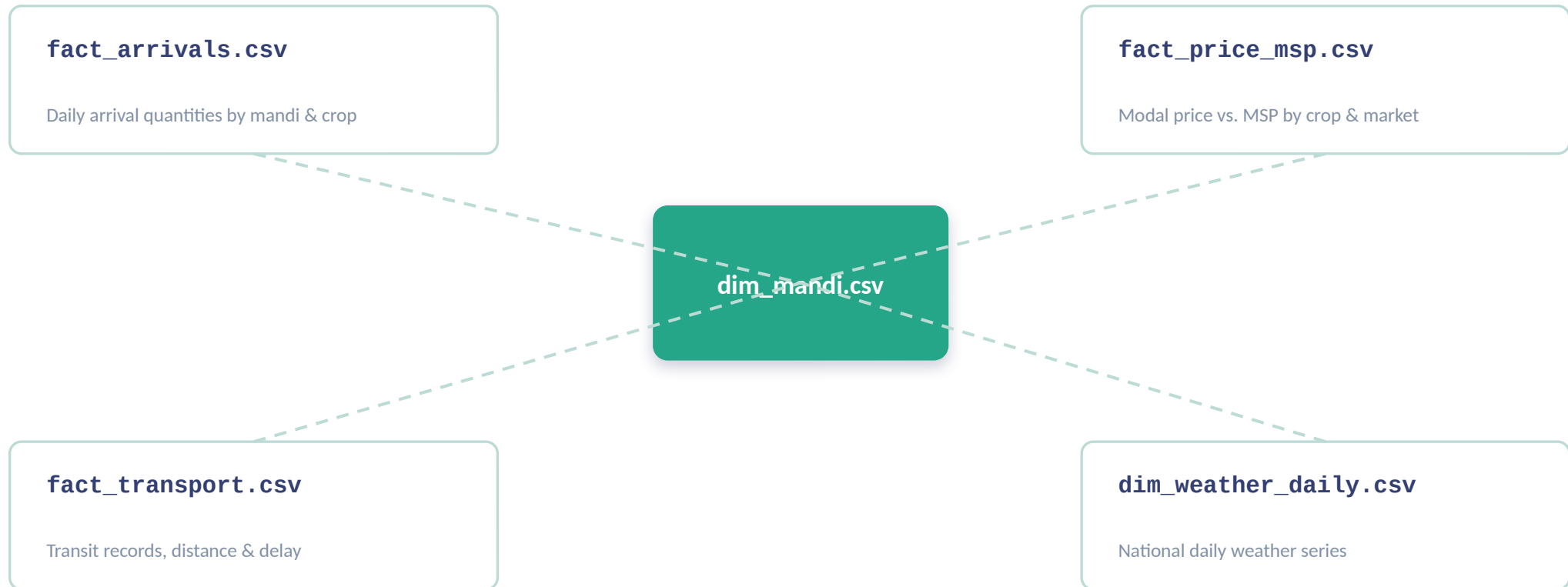
Key Cleaning Logic

Issue	Fix Applied
Mixed mandi ID formats (MANDI001, M017, 034...)	Normalized to canonical MANDI0## format
Crop names in English / Hindi / Punjabi, mixed case	Canonical mapping; raw value kept in raw_crop_name
Quantities in Tonnes / Quintals / KG, unit embedded	Parsed and converted to Quintals
Prices as ₹1,234 / Rs. 1,234 / INR 1234 / 1234/-	Symbols & commas stripped, converted to numeric
Negative arrival quantities	Converted to absolute value, flagged qty_was_negative
Missing farmer counts	Imputed with crop-level median, flagged
Weather in UTC/IST, °C/°F, mm/inches	Standardized to IST, Celsius, millimeters

Full reasoning for every decision lives in docs/data_dictionary.md and docs/cleaning_report.md.

Analytics Layer: Star Schema

Five clean CSVs form a star schema — one mandi dimension joined to four fact tables covering arrivals, price, transport and weather.



Working Flow

01

Collect Data

Pull the 5 raw source files from raw_data/

03

Model the Schema

Load into a star schema of 5 query-ready CSVs

05

Deploy the Agent

Ship the Streamlit text-to-chart AI layer

02

Clean & Standardize

Run clean_pipeline.py to fix formats and flag issues

04

Build the Dashboard

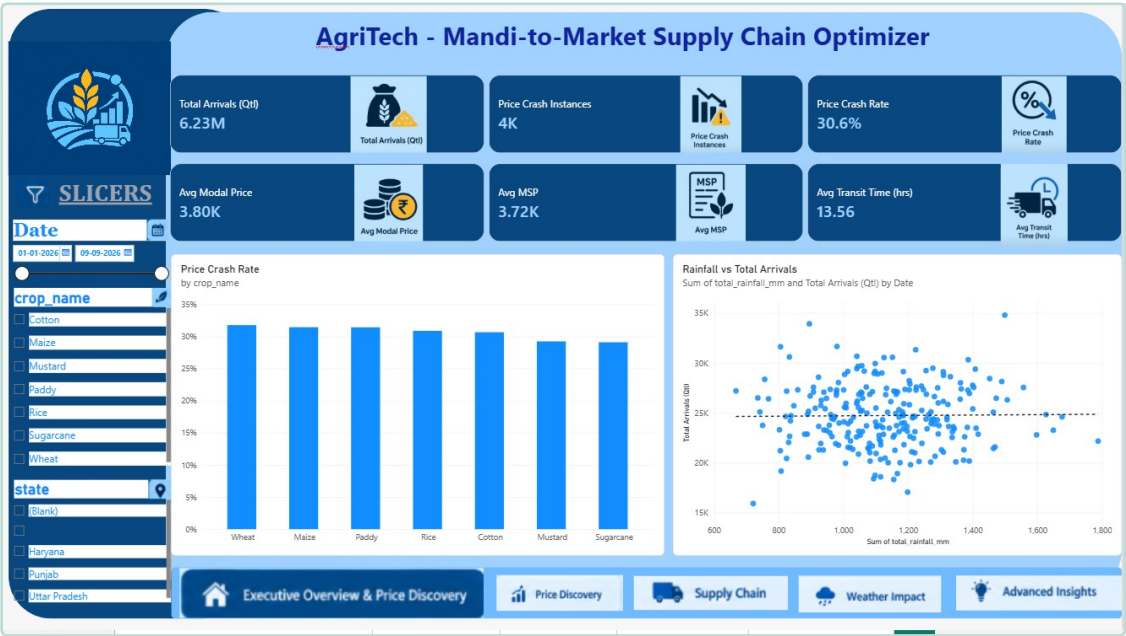
Design DAX measures and Power BI report pages

06

Publish & Share

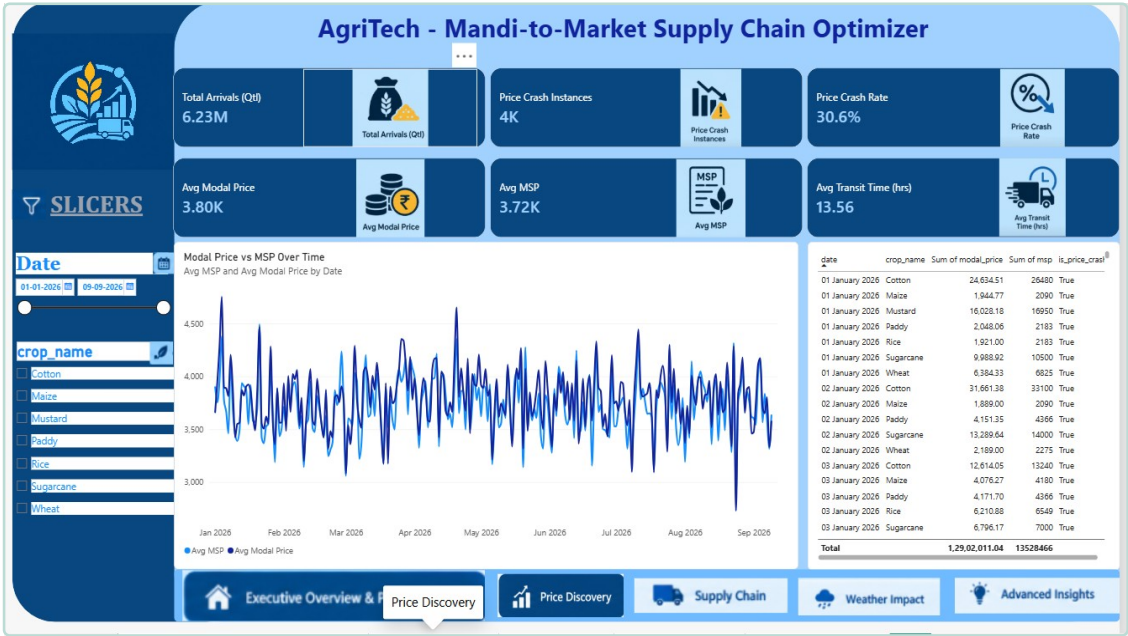
Push to GitHub and publish both live links

Dashboard: Executive Overview & Price Discovery



Executive Overview

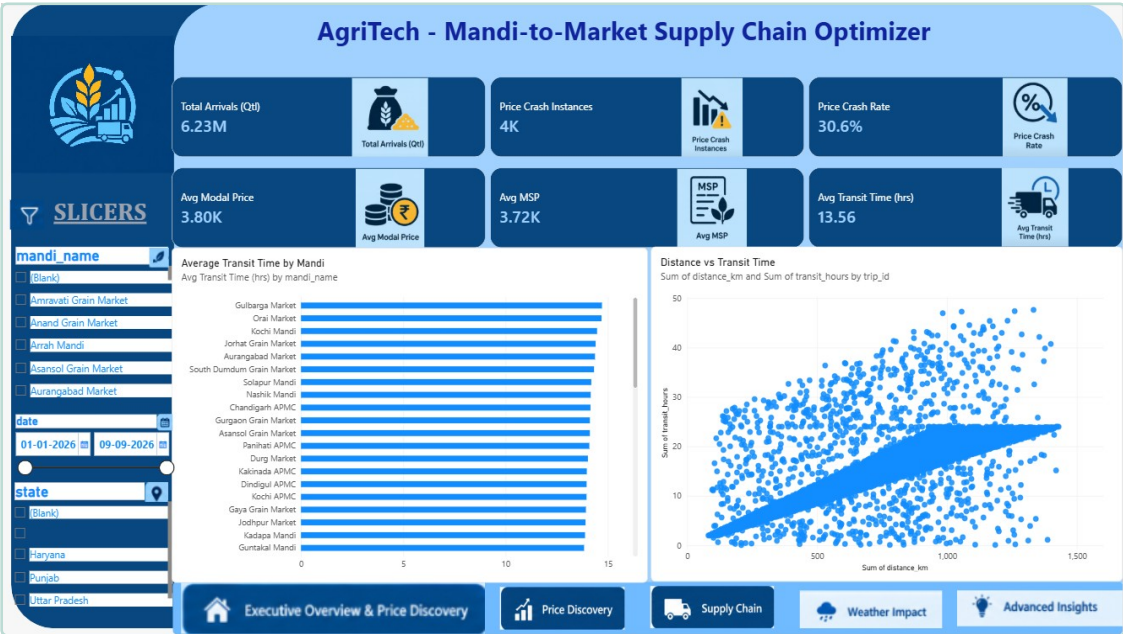
Total arrivals, top mandis by volume, and headline KPIs at a glance.



Price Discovery

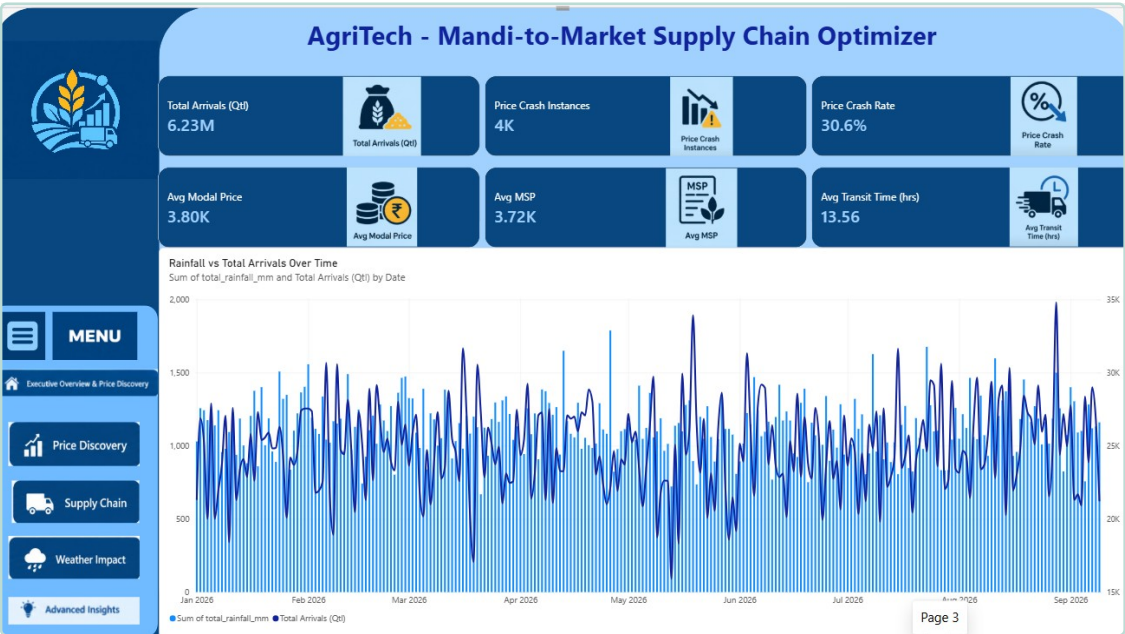
Modal price vs. MSP by crop, and Price-Crash instances flagged for review.

Dashboard: Supply Chain & Weather Impact



Supply Chain

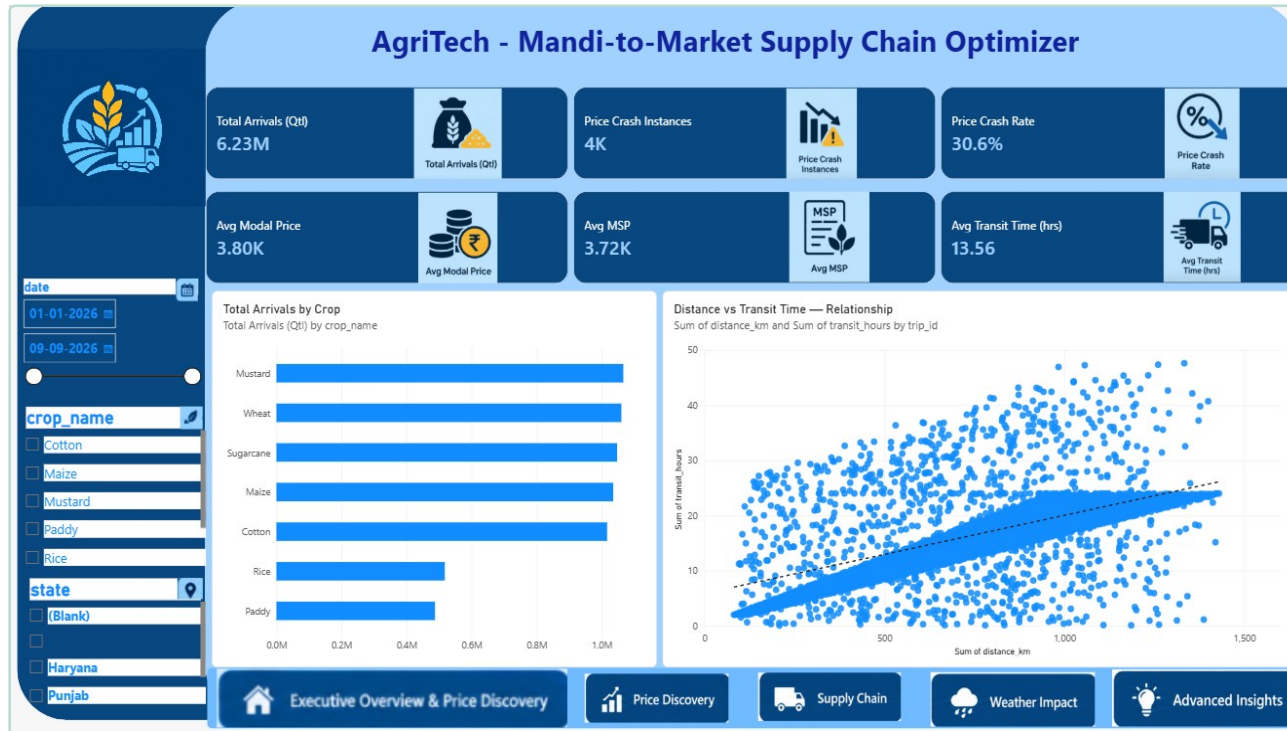
Transit time, delay rate, and route-level logistics performance across mandis.



Weather Impact

Rainfall and temperature trends layered against arrivals to surface weather-driven risk.

Advanced Insights & Bonus AI Agent



Advanced Insights page



Text-to-Chart AI Agent

Ask in plain English: "Plot the daily arrival trend of Wheat in Amritsar mandi vs MSP for the last 30 days"

- Picks the right chart type automatically
- Generates a one-line factual summary
- Built with Streamlit + Plotly
- Deployed live for judges to try instantly

Results & Performance

The pipeline turns fragmented inputs into a reliable analytics foundation, measured across three dimensions:

5 → 5

Raw files fully rescued into clean, query-ready CSVs

100%

Cleaning decisions documented & reproducible

2 Live

Deployments — Power BI dashboard & AI agent

Core metrics tracked: Total arrivals (Quintals), Modal Price vs. MSP, Price-Crash instances, Transit time & delay rate, Top mandis by volume.



Advantages



Fully Auditable

Every cleaning decision is traceable back to its documented reasoning.



Reproducible

Idempotent pipeline — rerun anytime, same clean output every time.



Scalable Schema

Star schema design extends cleanly to new seasons or crops.



Multilingual-Aware

Handles English, Hindi and Punjabi crop names without data loss.



Self-Serve Insight

The AI agent answers new questions without touching the pipeline.



Applications

This system can be used by:



Individual Farmers & Traders

Check price trends and MSP gaps before deciding where to sell.



Mandi Boards

Monitor arrivals and flag price crashes across markets in real time.



Agri-Tech Companies

Plug the clean CSVs into their own forecasting or logistics tools.

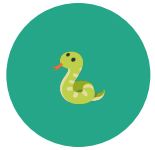


Government Agri Departments

Track transport delays and weather risk at a state or national level.



Tech Stack



Python

pandas, numpy, openpyxl — cleaning & transformation



Jupyter Notebook

Interactive, inspectable pipeline documentation



Power BI Desktop

Data modeling, DAX measures, dashboard design



Streamlit + Plotly

Text-to-chart AI agent, deployed live



Roadmap: Where This Goes Next

1

District-level weather mapping

Once sensor metadata is available, move past the national weather series.

2

Scheduled refresh

Automate cleaning + dashboard refresh instead of manual re-runs each cycle.

3

Year-over-year comparisons

Add trend comparisons for arrivals and prices as more seasons arrive.

4

A smarter agent

Expand the AI agent with follow-ups, forecasting, and anomaly detection.



Conclusion

This project turns fragmented agricultural data into a trustworthy, reusable analytics foundation — balancing speed, rigor and transparency throughout.



Data Rescue

A documented, reproducible pipeline turns 5 messy files into 5 trusted CSVs.



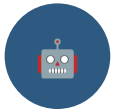
Analytics Layer

A star schema built for fast, reliable BI — not a one-off script.



Actionable Dashboard

5 Power BI pages surface the metrics that actually drive decisions.



Self-Serve AI

A bonus agent means new questions don't need a new pipeline run.



Submission-ready:
code, data dictionary,
dashboard link & PDF deck.

THANK YOU

We appreciate your time and attention.

This project represents a meaningful step toward applying data science to real-world agricultural supply-chain challenges — turning fragmented records into decisions people can trust.



Live Dashboard

Power BI report — Executive, Price, Supply Chain, Weather & Advanced Insights



Live AI Agent

Streamlit text-to-chart analytics agent, deployed and interactive



GitHub Repository

Full pipeline, data dictionary, cleaning report & Power BI file

github.com/rohansaii/AgriTech-Mandi-to-Market-Supply-Chain-Optimizer